

The Silent Tax: Typo-Robustness Analysis (MATH500)

DeepSeek-R1-Distill-Qwen-7B

All tables are *answered-only*: traces that hit the token cap before writing a final answer are excluded from the metrics. Configs are named `typo{rate}_real{ratio}`: *rate* = % of words corrupted, *real* = % of typos that form real words. Baseline is the clean input.

Accuracy and Flips

Table 1: Accuracy per config. `n` = questions; `n_answered` = questions that produced a final answer (not truncated at the token cap). `acc_answered` = correct / `n_answered`; `acc_strict` = correct / `n` (truncated counted as wrong); `ci_lo/ci_hi` = 95% bootstrap CI on `acc_strict`.

config	n	n_answered	acc_answered	acc_strict	ci_lo	ci_hi
clean	500	496	0.9516	0.944	0.924	0.964
typo25_real10	500	495	0.9394	0.93	0.906	0.95
typo25_real40	500	498	0.9337	0.93	0.906	0.952
typo25_real70	500	494	0.9393	0.928	0.904	0.948
typo50_real10	500	498	0.9277	0.924	0.9	0.946
typo50_real40	500	498	0.9137	0.91	0.884	0.934
typo50_real70	500	496	0.9153	0.908	0.882	0.932
typo75_real10	500	494	0.8907	0.88	0.85	0.908
typo75_real40	500	496	0.881	0.874	0.844	0.9
typo75_real70	500	497	0.8692	0.864	0.832	0.892

Table 2: Answered-only flips vs clean, paired by question over `n_both` (answered in both conditions). `r2w` / `w2r` = questions correct on clean but wrong on this config / wrong on clean but correct on this config; `r2w_ratio` = `r2w` / `n_both`, `w2r_ratio` = `w2r` / `n_both`; `net_flip_ratio` = `r2w_ratio` - `w2r_ratio` = (`r2w` - `w2r`) / `n_both`, the net rate at which this config worsens answers relative to clean.

config	n_both	r2w	w2r	r2w_ratio	w2r_ratio	net_flip_ratio	mcnemar_b	mcnemar_c	mcnemar_stat	mcnemar_p
typo25_real10	494	12	7	0.0243	0.0142	0.0101	12	7	0.842	0.35879536
typo25_real40	495	15	9	0.0303	0.0182	0.0121	15	9	1.042	0.30743417
typo25_real70	493	15	9	0.0304	0.0183	0.0122	15	9	1.042	0.30743417
typo50_real10	495	18	9	0.0364	0.0182	0.0182	18	9	2.37	0.12365771
typo50_real40	495	22	6	0.0444	0.0121	0.0323	22	6	8.036	0.00458639
typo50_real70	494	25	9	0.0506	0.0182	0.0324	25	9	6.618	0.01009731
typo75_real10	493	34	5	0.069	0.0101	0.0588	34	5	20.103	7.34e-06
typo75_real40	492	39	7	0.0793	0.0142	0.065	39	7	20.891	4.86e-06
typo75_real70	494	46	7	0.0931	0.0142	0.0789	46	7	27.245	1.8e-07

McNemar test (used in the flips table above).

- **What it is:** a paired significance test for before/after binary outcomes. It looks only at the questions that *changed* (the discordant pairs: *b* = right→wrong, *c* = wrong→right) and ignores those that stayed the same.

- **Null hypothesis:** typos break correct answers as often as they fix wrong ones ($b = c$, i.e. no net effect).
- **Statistic:** $(|b - c| - 1)^2 / (b + c)$ (column `mcnemar_stat`); `mcnemar_p` is its chi-square p -value with 1 degree of freedom.
- **When we reject it:** when `mcnemar_p` < 0.05 — the right→wrong vs wrong→right imbalance is too large to be chance.

Reasoning Length

Table 3: Absolute generated tokens per config (answered-only).

config	n_answered	mean_tok	median_tok	p90_tok	max_tok
clean	496	2926.7	2317.0	5689	14397
typo25_real10	495	2944.6	2354	5810	12417
typo25_real40	498	3110.7	2361.5	5990	16219
typo25_real70	494	2991.9	2261.5	5712	14082
typo50_real10	498	2938.9	2317.5	5174	15267
typo50_real40	498	3085.2	2349.0	5851	16547
typo50_real70	496	3086.9	2295.5	5908	14559
typo75_real10	494	2945.2	2350.0	5419	16258
typo75_real40	496	3125.0	2359.5	5986	15195
typo75_real70	497	3137.5	2352	5833	16050

Table 4: Mean generated tokens for correct vs wrong answers. `wrong_over_correct` = `tok_wrong` / `tok_correct` (mean tokens of wrong answers divided by mean tokens of correct answers).

config	tok_correct	tok_wrong	wrong_over_correct
clean	2803.4	5351.7	1.909
typo25_real10	2826.0	4783.0	1.6925
typo25_real40	2900.4	6074.8	2.0945
typo25_real70	2859.1	5046.8	1.7652
typo50_real10	2763.4	5191.0	1.8785
typo50_real40	2838.6	5694.0	2.0059
typo50_real70	2854.5	5599.5	1.9617
typo75_real10	2786.8	4235.8	1.52
typo75_real40	2881.0	4931.9	1.7118
typo75_real70	2911.5	4639.5	1.5935

Self-doubt

The self-doubt markers are curated words grouped into two *banks*, each matched as a case-insensitive whole-word pattern in the reasoning text. Density is matches per 1000 reasoning words.

- **second_guess** — the proposal’s canonical self-correction words plus direct reversals: *wait, hmm, actually, reconsider, but wait, hold on, on second thought, let me recheck, double-check*.
- **uncertainty** — hedging / “not certain” words: *maybe, perhaps, possibly, might be, could be, not sure, I guess, presumably, or maybe / alternatively, unclear / confused*.

Example reasoning: “*Wait, maybe the total is 12... hmm, actually let me recheck*” contributes *wait*, *hmm*, *actually* (second_guess) and *maybe* (uncertainty). A separate *typo-noticing* bank (*typo*, *misspelled*, *doesn't make sense*, ...) is treated as repair behaviour, not self-doubt.

Table 5: Doubt-marker density per config: second_guess and uncertainty, per 1k words and per trace.

config	n_answered	second_guess_per_1k	second_guess_per_trace	uncertainty_per_1k	uncertainty_per_trace	total_per_1k	total_per_trace
clean	496	12.558	17.667	6.754	9.502	19.312	27.169
typo25_real10	495	11.818	16.794	6.969	9.903	18.787	26.697
typo25_real40	498	13.722	20.655	7.5	11.289	21.223	31.944
typo25_real70	494	13.172	18.96	8.238	11.858	21.41	30.818
typo50_real10	498	12.694	18.213	7.934	11.384	20.629	29.596
typo50_real40	498	12.747	19.247	8.756	13.221	21.503	32.468
typo50_real70	496	13.136	19.772	8.48	12.764	21.616	32.536
typo75_real10	494	12.687	17.984	8.079	11.451	20.766	29.435
typo75_real40	496	12.705	19.482	9.027	13.843	21.732	33.325
typo75_real70	497	12.549	19.394	9.326	14.414	21.875	33.809

Table 6: Per-marker density (markers per 1000 reasoning words): clean vs pooled-typo. discrimination = typo minus clean (larger = more typo-responsive). All bank markers, ranked by discrimination.

marker	category	clean_per_1k	typo_per_1k	discrimination
maybe	uncertainty	2.848	3.793	0.946
wait	second_guess	6.535	6.983	0.448
or maybe/alt	uncertainty	1.966	2.191	0.224
perhaps	uncertainty	1.274	1.486	0.212
unclear/confused	uncertainty	0.132	0.21	0.079
hmm	second_guess	2.114	2.15	0.037
might be	uncertainty	0.188	0.219	0.031
could be	uncertainty	0.1	0.113	0.013
not entirely	uncertainty	0.006	0.015	0.009
not sure	uncertainty	0.168	0.174	0.006
possibly	uncertainty	0.001	0.005	0.003
hold on	second_guess	0.639	0.641	0.002
presumably	uncertainty	0.0	0.001	0.001
I assume	uncertainty	0.01	0.01	0.0
on second thought	second_guess	0.0	0.0	0.0
reconsider	second_guess	0.003	0.002	-0.0
I guess	uncertainty	0.062	0.057	-0.005
wait/actually no	second_guess	0.603	0.571	-0.032
actually	second_guess	0.645	0.607	-0.038
double-check	second_guess	0.514	0.472	-0.043
but wait	second_guess	0.837	0.78	-0.057
let me recheck	second_guess	0.668	0.601	-0.067

Table 7: Doubt density (markers per 1000 reasoning words) split by final outcome, per category (2g = second.guess, un = uncertainty, tot = both). For a group, density = total markers in that group / total reasoning words in that group x 1000. *_correct = over correct-answer traces; *_wrong = over wrong-answer traces; *_wrong_over_correct = _wrong / _correct.

config	2g_correct	2g_wrong	2g_wrong_over_correct	un_correct	un_wrong	un_wrong_over_correct	tot_correct	tot_wrong	tot_wrong_over_correct
clean	12.356	14.447	1.169	6.327	10.735	1.697	18.683	25.182	1.348
typo25_real10	11.614	13.593	1.17	6.543	10.663	1.63	18.157	24.256	1.336
typo25_real40	13.011	17.929	1.378	6.978	10.588	1.517	19.989	28.516	1.427
typo25_real70	12.744	16.619	1.304	7.667	12.846	1.676	20.411	29.465	1.444
typo50_real10	12.079	16.563	1.371	7.004	13.787	1.968	19.083	30.35	1.59
typo50_real40	12.348	14.628	1.185	7.496	14.696	1.961	19.844	29.323	1.478
typo50_real70	12.277	17.265	1.406	7.344	13.938	1.898	19.62	31.203	1.59
typo75_real10	11.849	16.775	1.416	6.839	14.124	2.065	18.688	30.899	1.653
typo75_real40	11.786	16.224	1.377	7.405	15.24	2.058	19.191	31.464	1.639
typo75_real70	11.772	15.448	1.312	8.054	14.074	1.747	19.826	29.522	1.489

LLM judge. The proposal specifies a second, non-lexical measure for both dimensions: a separate model with a fixed, strict prompt. The judge is a different model from the one under test, called at temperature 0 on a seeded random subsample of answered traces per config. It is shown the corrupted-word list (*original* $\rightarrow$ *as shown*), the question, and the reasoning trace (head+tail window). It returns two scalar ratings and evidence: *self_doubt_score* from 0–10, where 0 means no second-guessing and 10 means pervasive uncertainty or looping; and *repair_understanding_score* from 0–5, where 0 means the question’s meaning is fully understood and 5 means the model is fundamentally lost because of the corruption. The judge also returns a one-line justification and verbatim evidence quotes; quotes are checked against the reasoning trace. The judge is a subsample measure, while the lexical tables remain the full-data result.

Table 8: New scalar LLM-judge analysis per config. mean/median *self_doubt_score* use the 0-10 scale; mean/median *repair_understanding_score* use the 0-5 scale; threshold columns show the share of traces with substantial doubt (≥ 5) or substantial loss of meaning (≥ 3).

config	n	mean_self_doubt	median_self_doubt	frac_self_doubt_ge5	mean_repair_understanding	median_repair_understanding	frac_repair_understanding_ge3	evidence_self_doubt_frac	evidence_repair_frac
clean	464	3.657	2	0.3513	0.032	0	0.0043	0.9073	0.9914
typo25_real10	463	3.767	2	0.3931	0.257	0	0.0108	0.9114	0.8963
typo25_real40	447	3.964	3	0.4273	0.36	0	0.0268	0.9016	0.8881
typo25_real70	455	3.958	2	0.4308	0.433	0	0.0308	0.8901	0.9011
typo50_real10	460	4.063	4	0.4413	0.322	0	0.0304	0.913	0.9065
typo50_real40	446	3.996	2	0.4126	0.365	0	0.0426	0.9126	0.9215
typo50_real70	454	3.888	2	0.4075	0.366	0	0.0286	0.8855	0.9119
typo75_real10	456	3.77	2	0.3904	0.377	0	0.0395	0.9211	0.9167
typo75_real40	459	4.118	4	0.4553	0.442	0	0.0501	0.8954	0.9107
typo75_real70	460	4.133	3	0.4413	0.62	0	0.087	0.8804	0.8804

Table 9: New scalar LLM-judge scores split by final answer outcome. The correct/wrong columns are means over traces whose final answer was correct or wrong.

config	n	n_correct	n_wrong	mean_doubt_correct	mean_doubt_wrong	mean_repair_correct	mean_repair_wrong
clean	464	441	23	3.587	5.0	0.011	0.435
typo25_real10	463	434	29	3.673	5.172	0.212	0.931
typo25_real40	447	419	28	3.84	5.821	0.298	1.286
typo25_real70	455	427	28	3.848	5.643	0.377	1.286
typo50_real10	460	429	31	3.96	5.484	0.266	1.097
typo50_real40	446	406	40	3.786	6.125	0.259	1.45
typo50_real70	454	415	39	3.636	6.564	0.265	1.436
typo75_real10	456	403	53	3.541	5.509	0.211	1.642
typo75_real40	459	404	55	3.886	5.818	0.334	1.236
typo75_real70	460	396	64	3.801	6.188	0.475	1.516
all	4564	4174	390	3.755	5.808	0.268	1.305

Table 10: New scalar LLM-judge scores pooled by real-word typo ratio, showing whether higher real-word corruption is associated with more doubt or loss of understanding.

real_ratio	n	mean_self_doubt	mean_repair_understanding	frac_self_doubt_ge5	frac_repair_understanding_ge3
10	1379	3.867		0.318	0.4083
40	1352	4.027		0.39	0.432
70	1369	3.993		0.473	0.4266

Repair Behaviour

Table 11: How each corrupted word was handled, per config (categories defined below the table).

config	n_corrupt_words	silent_fix_n	flagged_n	misread_n	not_used_n	silent_fix_frac	flagged_frac	misread_frac	not_used_frac
typo25_real10	1882	1468	44	23	347	0.78	0.0234	0.0122	0.1844
typo25_real40	1901	1389	110	31	371	0.7307	0.0579	0.0163	0.1952
typo25_real70	1881	1342	192	32	315	0.7135	0.1021	0.017	0.1675
typo50_real10	3661	2744	157	51	709	0.7495	0.0429	0.0139	0.1937
typo50_real40	3661	2569	305	85	702	0.7017	0.0833	0.0232	0.1918
typo50_real70	3651	2452	414	107	678	0.6716	0.1134	0.0293	0.1857
typo75_real10	5282	3720	433	111	1018	0.7043	0.082	0.021	0.1927
typo75_real40	5260	3600	504	128	1028	0.6844	0.0958	0.0243	0.1954
typo75_real70	5297	3296	805	214	982	0.6222	0.152	0.0404	0.1854

A *corrupted word* is a word that received a typo — it differs between the clean and typo question (e.g. *sum* → *sun*). We recover these by diffing the two questions, then place each into one bucket by which form appears in the reasoning:

- **silent_fix** — only the original word (*sum*) appears: a quiet recovery.
- **flagged** — both forms appear (*sum* and *sun*): noticed and fixed.
- **misread** — only the corrupted form (*sun*) appears: read as the wrong word.
- **not_used** — neither form appears in the reasoning.

Columns ***_n** are counts; ***_frac** are the share of **n_corrupt_words**.

Table 12: Word handling pooled by real-word ratio: the non-word vs real-word contrast.

real_ratio	n_corrupt_words	silent_fix_frac	flagged_frac	misread_frac	not_used_frac
10	10825	0.7327	0.0586	0.0171	0.1916
40	10822	0.6984	0.0849	0.0225	0.1941
70	10829	0.6547	0.1303	0.0326	0.1824

Table 13: Misread rate split by final correctness. A typo’d word is ‘misread’ when the reasoning uses only its corrupted form (e.g. sun instead of sum). Within each outcome group, misread rate = (typo’d words that were misread) / (all typo’d words examined), pooled over the group. misread_correct = over problems answered correctly; misread_wrong = over problems answered wrong; wrong_over_correct = misread_wrong / misread_correct.

config	misread_correct	misread_wrong	wrong_over_correct
typo25_real10	0.0133	0.0044	0.331
typo25_real40	0.0132	0.0388	2.943
typo25_real70	0.0138	0.0411	2.97
typo50_real10	0.0125	0.0254	2.022
typo50_real40	0.0217	0.0322	1.483
typo50_real70	0.024	0.0604	2.513
typo75_real10	0.017	0.037	2.175
typo75_real40	0.0187	0.0476	2.547
typo75_real70	0.0345	0.0604	1.751

Real-word Effect

This section isolates the effect of typo *kind* (real-word vs non-word), holding the question, corrupted positions, and rate fixed — something the vs-clean accuracy section cannot do. Three views:

- a **controlled paired** comparison of low-real vs high-real variants on the *same* questions, with a McNemar test (paired table);
- a **per-typo logistic regression** comparing the marginal harm of one real-word vs one non-word typo (logit table);
- whether wrong answers **flag the corruption** more or less as the real-word share rises (silent-failure table).

Read the signs, deltas and p -values in the tables below for this dataset’s result.

Table 14: Controlled paired comparison of two TYPO variants against EACH OTHER (baseline is the low-real variant, NOT clean – so this is not the vs-clean accuracy table). Within a fixed rate the same questions are corrupted in the same positions, only the typo KIND differs. Restricted to questions answered in both variants. acc_low_real / acc_high_real = accuracy on the lower / higher real-word variant; delta_acc = acc_high_real - acc_low_real; r2w = questions correct in the low-real variant but wrong in the high-real one; w2r = the reverse; mcnemar_p = McNemar test on that paired change. Negative delta with r2w $\hat{>}$ w2r means real-word typos are more harmful than the non-word typos they replaced.

rate	compare	n_both	acc_low_real	acc_high_real	delta_acc	r2w	w2r	mcnemar_p
25	real10_to_real70	492	0.9411	0.9411	0.0	10	10	0.82306327
25	real10_to_real40	494	0.9393	0.9393	0.0	10	10	0.82306327
25	real40_to_real70	493	0.9412	0.9391	-0.002	13	12	1.0
50	real10_to_real70	496	0.9294	0.9153	-0.0141	19	12	0.2811981
50	real10_to_real40	498	0.9277	0.9137	-0.0141	17	10	0.24821308
50	real40_to_real70	496	0.9173	0.9153	-0.002	13	12	1.0
75	real10_to_real70	494	0.8907	0.8725	-0.0182	28	19	0.24324271
75	real10_to_real40	490	0.8898	0.8878	-0.002	20	19	1.0
75	real40_to_real70	493	0.8844	0.8682	-0.0162	25	17	0.28008721

Table 15: Logistic regression $\text{correct} \sim \text{num_real} + \text{num_nonword}$ over all answered typo traces ($\text{num_real} / \text{num_nonword}$ = count of real-word / non-word typos in the problem). coef_* = change in log-odds of a correct answer per one added typo of that kind (more negative = more harmful); $\text{odds}_* = \exp(\text{coef})$; intercept is the model constant. n = number of traces fit.

n	coef_num_real	coef_num_nonword	odds_real	odds_nonword	intercept
4466	-0.07605	-0.05882	0.9268	0.9429	3.01851

Table 16: Among WRONG answers only, does the model flag the corruption, by real ratio? $\text{explicit_notice_frac}$ = fraction of wrong-answer traces whose reasoning uses strong typo-flagging language (typo / misspelled / doesn’t make sense / ...); $\text{silent_frac} = 1 - \text{explicit_notice_frac}$. The proposal predicts real-word failures are quieter ($\text{explicit_notice_frac}$ should fall as real rises).

real_ratio	n_wrong	explicit_notice_frac	silent_frac
10	120	0.45	0.55
40	135	0.5037	0.4963
70	137	0.4964	0.5036

Lexical Grid (overview)

Table 17: All four marker families (second_guess , uncertainty , typo_noticing , repair_words) across configs.

config	second_guess	uncertainty	typo_noticing	repair_words
clean	12.558	6.754	1.102	0.775
typo25_real10	11.818	6.969	1.461	0.884
typo25_real40	13.722	7.5	1.278	0.891
typo25_real70	13.172	8.238	1.433	0.962
typo50_real10	12.694	7.934	1.39	0.908
typo50_real40	12.747	8.756	1.426	0.95
typo50_real70	13.136	8.48	1.515	0.959
typo75_real10	12.687	8.079	1.542	1.061
typo75_real40	12.705	9.027	1.603	1.064
typo75_real70	12.549	9.326	1.591	1.113

Spellcheck Recovery